COMPLEXES WITH COHOMOLOGY IN A WEAK SERRE SUBCATEGORY FORM A SATURATED TRIANGULATED SUBCATEGORY

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6161-6177, following the statement at 6152-6159.

Editorial note. No omissions: the summand and triangle cases are argued; bounded versions follow the established mirror counting practice.

Allowed standard prerequisites (editorial).

- Triangulated subcategory criterion. lemma-triangulated-subcategory (same chapter, cited at ~6167). Reduces to shift preservation and triangle closure.
- Weak Serre subcategory. Homology, Definition homology-definition-serre-subcategory (cross-chapter, cited at ~6172). Kernels/cokernels bookkeeping for H^n(Z).

Source reference. Original equation 05ST: sources/stacks-derived/pinned-original-source.tex, source lines 3857–3857. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 02MO: focused-seeds/geometry/S9-der-cohomology-in-serre-subcategory/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05QX: sources/stacks-derived/pinned-original-source.tex, source lines 1077–1077. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $\mathcal{B} \subset \mathcal{A}$ be a weak Serre subcategory. The category $D_{\mathcal{B}}(\mathcal{A})$ is a strictly full saturated triangulated subcategory of $D(\mathcal{A})$. Similarly for the bounded versions.*

Proof. It is clear that $D_{\mathcal{B}}(\mathcal{A})$ is an additive subcategory preserved under the translation functors. If $X \oplus Y$ is in $D_{\mathcal{B}}(\mathcal{A})$, then both $H^n(X)$ and $H^n(Y)$ are kernels of maps between maps of objects of $\mathcal{B}$ as $H^n(X \oplus Y) = H^n(X) \oplus H^n(Y)$. Hence both X and Y are in $D_{\mathcal{B}}(\mathcal{A})$. By Lemma 05QX it therefore suffices to show that given a distinguished triangle (X, Y, Z, f, g, h) such that X and Y are in $D_{\mathcal{B}}(\mathcal{A})$

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then Z is an object of $D_{\mathcal{B}}(\mathcal{A})$. The long exact cohomology sequence (05ST) and the definition of a weak Serre subcategory (see Homology, Definition 02MO) show that $H^n(Z)$ is an object of $\mathcal{B}$ for all n. Thus Z is an object of $D_{\mathcal{B}}(\mathcal{A})$. $\square$